INCH-POUND

MIL-C-70462C (AR)
AMENDMENT 5
13 May 1992
SUPERSEDING
AMENDMENT 4
28 May 1991

MILITARY SPECIFICATION

CHARGE, PROPELLING, 155MM, M203A1
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-C-70462C (AR) dated 4 April 1986, and is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of Department of Defense.

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2.1.2 Add the following new drawing:

"9345091 Case Cartridge Assembly 155MM M203A1"

3.3.1 Delete in its entirety and substitute the following new paragraph:

"3.3.1 Propellant weight. The propellant weight shall comply with the recommended charge weight specified in the loading authorization within plus or minus 0.25 ounce. The loading authorization for the recommended charge weight as specified in the Propellant Acceptance Sheet may require weight adjustment if the manufacturer of the combustible case components for the propelling charge loading, assembling and packing is different from the manufacturer of the combustible cases for calibration components which were used for the propellant acceptance test. The weight adjustment for the recommended charge weight shall be obtained from the first article test for the new contractor for manufacturing of the combustible cases for the propelling charge as specified in MIL-C-70461."

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3.5 Delete in its entirety and substitute new paragraph 3.5:

"3.5 Moisture content of black powder. The moisture content of the black powder at the igniter loading station at the time of loading shall not exceed 0.7%. The moisture content of the black powder in the basepad of the propelling charge shall be determined in accordance with 4.5.5.

3.6 Delete entire paragraph.

3.8 Delete heading "Avg Press Max (PSI)" and all entries under the same heading.

* Add the following new paragraph:

"3.10.1 First article test. Test samples shall be withdrawn from the first lot produced by the contractor and tested in accordance with all the requirements of this specification and with Table IV. Until the first article test had been performed and found to be acceptable, the contractor is in no way authorized by the Government to initiate second lot production unless otherwise directed by the contracting officer."

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* 4.4.1a Add the following to the end of the paragraph 4.4.1a:

"Entire lot of propellant shall be utilized in the formation of one lot of propelling charge unless otherwise approved by the contracting officer."

4.4.1d Delete in its entirety and substitute the following new paragraphs:

"4.4.1d Case bodies from not more than one interfix number from one manufacturer.

4.4.1e Igniter containers and rear caps from not more than one interfix number from one manufacturer."

4.4.1d Delete in its entirety and substitute the following new paragraph:

"4.4.1.d Case cartridge assembly (Dwg #9345091) from not more than one interfix number from one manufacturer."

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4.4.2.7 Major 102: Delete "completely" and substitute "securely".

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4.4.2.7 Delete Major Defect 105.

4.4.2.9 Add the following new major category:

"104	Bottom spacer missing (only when utilizing wrapped body assembly (Dwg. 12598522))".	100%	3.2	Visual

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4.5.6 Delete entire paragraph.

4.5.8 Delete in its entirety and substitute the following new paragraphs:

"4.5.8 Cloth, spun viscose rayon resin impregnated.

4.5.8.1 Cloth for base end igniter. Each roll, as a minimum, shall be tested for compliance with the physical characteristics listed in Table I in 3.5 of MIL-C-43157. Testing will be in accordance with 4.3.1 of MIL-C-43157. Any sample which fails to comply with the applicable requirement shall be cause for rejection of the cloth bolt represented by the sample. The method of inspection to be used during production is to be established by the contractor and approved by the Government. The method, as minimum, shall contain positive assurance that the material being utilized has been verified for compliance with the specification requirements as stated above (see 6.16).

4.5.8.2 Cloth for lead laminated rayon cloth, liner backing and strap. At the time cloth is introduced to the sewing or cutting operation, an identification will be made for each roll to verify that proper material is used. Any cloth failing to be identified as proper material in accordance with applicable drawing and specifications shall be removed from the sewing and cutting operation."

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4.5.9 Delete in its entirety and substitute new paragraph 4.5.9:

"4.5.9 Proving ground assessments. These tests shall be conducted at a Government proving ground in accordance with the applicable acceptance test procedure using M549, RA or PXR-6325 projectiles inert loaded within 0.1 pound of an as fired weight of 96.0 pounds using primer, M82 and fired from M198 Howitzer with cannon M199 with 75 percent (minimum) remaining life. Firing shall be conducted with a spindle to charge base standoff of 1 inch (plus or minus 1/4 inch). Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that when fired, the charges are at the conditioning temperature. The following tolerance shall apply to all 145 degrees F and 70 degrees F conditioning: 145 degrees F from 140 degrees F to 145 degrees F; 70 degrees F from 68 degrees F to 72 degrees F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. One rear piezo gage shall be used to measure the maximum charge pressure of each charge for information and ignition delay of each charge. Velocity coils will be used to measure the muzzle velocity of each charge. Temperature indicator (thermometer) strips shall be placed on the side of all charges or on the side of the metal containers, as applicable. If any indicator strip shows the charge has been exposed to a temperature greater than 145 degrees F, that charge will not be fired."

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4.5.9.3.2 Delete entire paragraph.

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4.5.11 Delete entire paragraph and substitute following new paragraph:

"4.5.11 Air pressure test. Each container shall be tested as specified on drawing 9354642. A flow rate (see 6.22) greater than the required specified requirement shall be cause for rejection."

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6.8 Delete heading "Avg Press Max" and all entries under the same heading.

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Add the following new paragraph:

"6.22 Flow rate method. The flow rate method is defined as a procedure to measure the passage of air or gas at ambient temperature from the container in a specified time."

* Add the following new paragraph:

"6.23 First article test. The first article test is a one time qualification test for a new contractor to load, assemble and pack acceptable M203A1 propelling charges. Once a contractor has demonstrated the ability to load, assemble and pack the M203A1 propelling charges, the first article test may be waived by the contracting officer. The following test program shall be used for the first article test:

Table IV
First article test

Test Phase	Temp (deg F)	No. Control Rounds	No. Test Rounds	TECOM TOP
Propellant Checkout	70		10	TOP-4-2-504
	145		10	TOP-4-2-504
	-60		10	TOP-4-2-504
Sequential Environmental	145	15	60	TOP-4-2-504
	-60	15	60	TOP-4-2-504"

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